

An overview of force measurement: sensing methods, multi-axis measurement, dynamic and static measurement, force estimation and applications.

Abdulrahman Albakri, and Ghada Haj Ayoub,

Abstract—The abstract goes here.

Index Terms—Force, Force Measurement.

I. INTRODUCTION

THIS demo file is intended to serve as a “starter file” for IEEE journal papers produced under L^AT_EX using IEEEtran.cls version 1.8b and later. I wish you the best of success.

mds

August 26, 2015

To be written and filled at the last stage

II. FORCE SENSING METHODES

A. Resistive force transducer

B. Capcitive force transducer

It is a device that converts capacitance changes into electrical signals. Parallel-flat capacitors are common in capacitive force sensing systems. The capacitance of a parallel-plate capacitor is given by (1)

$$C = \epsilon_0 \epsilon_r \frac{A}{d} \quad (1)$$

where ϵ_0 and ϵ_r are the permittivity of free space and dielectric material forming the gap respectively and $d \ll A$. When a mechanical force is applied to a planar capacitive transducer, the electrical capacitance changes. This change is because of the variation one or more of the parameters present in the relationship (1). To measure the change in capacitance, an oscillator circuit, switched-capacitor circuits or capacitive ac-bridges is used [1] [2]. Capacitive force transducers are inserted in MEMS Force sensors, which enable mass production [3]. Fig(1) shows a schematic drawing of two-axial MEMS capacitive force device. Capacitive force transducers are capable of measuring forces with mN even pN resolution in wide measurement ranges [4] [5]. Capacitive sensing elements can be made in micromachined silicon technology as well as in conventional nonsilicon technology [6]. In the literature, Rivadeneyra et al.[7] conduct a review study on the latest developments in printed capacitive sensors. The paper covered the key technologies used in printed electronics and the main

types of capacitive sensors fabricated using various materials and techniques. Kim et al.[8] designed a capacitive sensor for use in robotics. They use a plastic structure and insulating rubber capacitors. Due to its simple design, it can be manufactured at a lower cost than sensors currently available on the market. Pu et al.[9] presented a novel analytical model to improve the estimation accuracy of elastomer displacements for Y-type elastomer capacitive six-axis F/T sensor. The results showed the maximum relative error is 8.12%. Compared with the analytical model of previous method, these obtained analytical model is confirmed to be accurate and higher efficiency. The proposed analytical model is expected to provide a specific theoretical model of estimation, design and optimization for capacitive six-axis F/T sensors.

C. Inductive force transducer

D. Piezoelectric force transducer

E. Fiber optic force transducer

F. Vibrating wire force transducer

III. STATIC VS DYNAMIC FORCE MEASUREMENT

IV. MULTI-DIMENSIONAL FORCE MEASUREMENT

V. FORCE ESTIMATION

VI. FORCE MEASUREMENT APPLICATIONS

A. Force measurement in monitoring and testing tasks

B. Use of force measurement in automated systems

C. Force measurement in research and development laboratories

VII. CONCLUSION

The conclusion goes here.

REFERENCES

- [1] Q. Liang, D. Zhang, G. Coppola, Y. Wang, S. Wei, and Y. Ge, “Multi-dimensional mems/micro sensor for force and moment sensing: A review,” *IEEE Sensors Journal*, vol. 14, no. 8, pp. 2643–2657, 2014.
- [2] R. A. Brookhuis, R. G. P. Sanders, B. N. A. Sikkens, and R. J. Wiegerink, “Three-axis force-torque sensor with fully differential capacitive readout,” in *2016 IEEE 29th International Conference on Micro Electro Mechanical Systems (MEMS)*, pp. 885–888, IEEE, 2016.

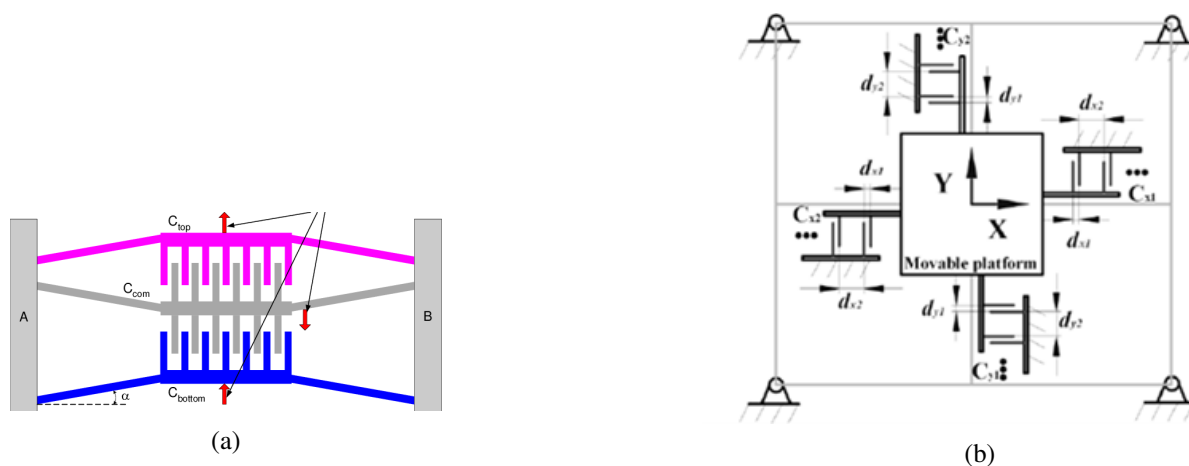


Fig. 1: (a) Capacitive force transducer [10], (b) Schematic representation of a two-axial MEMS capacitive force sensor [1]

- [3] F. Beyeler, S. Muntwyler, and B. Nelson, "Design and calibration of a microfabricated 6-axis force-torque sensor for microrobotic applications," vol. 1, pp. 520–525, 05 2009.
- [4] J. Fernandes and H. Jiang, "Three-axis capacitive touch-force sensor for clinical breast examination simulators," *IEEE Sensors Journal*, vol. 17, pp. 7231–7238, 11 2017.
- [5] D.-H. Lee, U. Kim, H. Jung, and H. R. Choi, "A capacitive-type novel six-axis force/torque sensor for robotic applications," *IEEE Sensors Journal*, vol. 16, no. 8, pp. 2290–2299, 2016.
- [6] X. Li and G. C. M. Meijer, *Capacitive Sensors*, ch. 8, pp. 225–248. John Wiley & Sons, Ltd, 2008.
- [7] A. Rivadeneyra and J. A. López-Villanueva, "Recent advances in printed capacitive sensors," *Micromachines*, vol. 11, no. 4, p. 367, 2020.
- [8] D. Kim, C. H. Lee, B. C. Kim, D. H. Lee, H. S. Lee, C. T. Nguyen, U. K. Kim, T. D. Nguyen, H. Moon, J. C. Koo, J. do Nam, and H. R. Choi, "Six-axis capacitive force/torque sensor based on dielectric elastomer," in *Electroactive Polymer Actuators and Devices (EAPAD) 2013* (Y. Bar-Cohen, ed.), vol. 8687, p. 86872J, International Society for Optics and Photonics, SPIE, 2013.
- [9] M. Pu, Q. Luo, Q. Liang, and J. Zhang, "Modeling for elastomer displacement analysis of capacitive six-axis force/torque sensor," *IEEE Sensors Journal*, vol. 22, no. 2, pp. 1356–1365, 2022.
- [10] M. Suster, N. Chaimanonart, and D. Young, "Low-noise cmos integrated sensing electronics for capacitive mems strain sensors," pp. 693 – 696, 11 2004.



Michael Shell Biography text here.

John Doe Biography text here.

Jane Doe Biography text here.